

# ***Business Continuity Planning with Bareos and rear***

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# Who am I ?

- **Independent Unix System Engineer since 1996**
- **Unix user since 1986**
- **Linux user since 1991**
- **Open Source contributor:**
  - **Upgrade-UX**
  - **Relax and Recover (rear)**
  - **SIM Installation and Logging (WBEMextras)**
  - **Adhoc Copy and Run (adhocr)**
  - **Config-to-HTML (cfg2html v6.x)**
  - **Make CD-ROM Recovery (mkCDrec)**

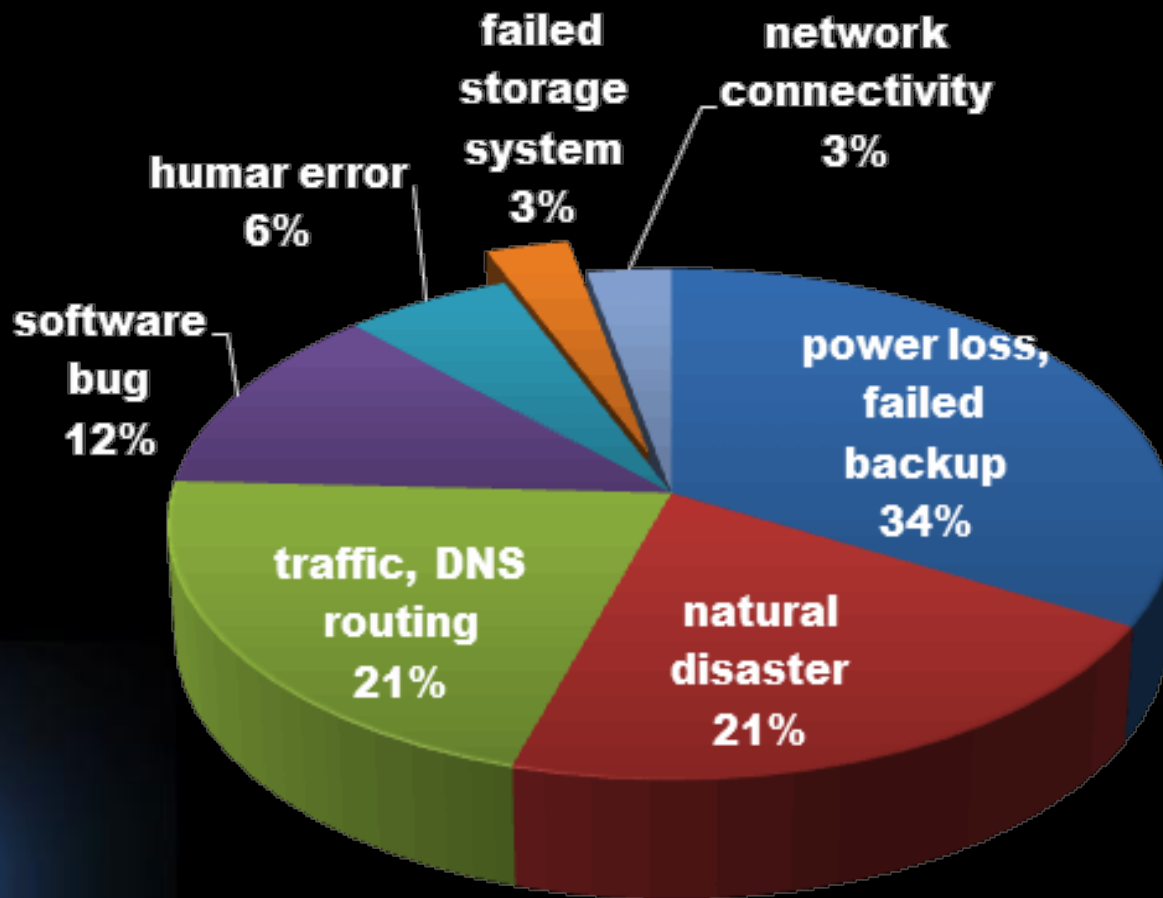
# Buzz Words

**Backup and Replication**

**Cloud disaster recovery**

**Business Continuity**

# Types of Business Interruptions



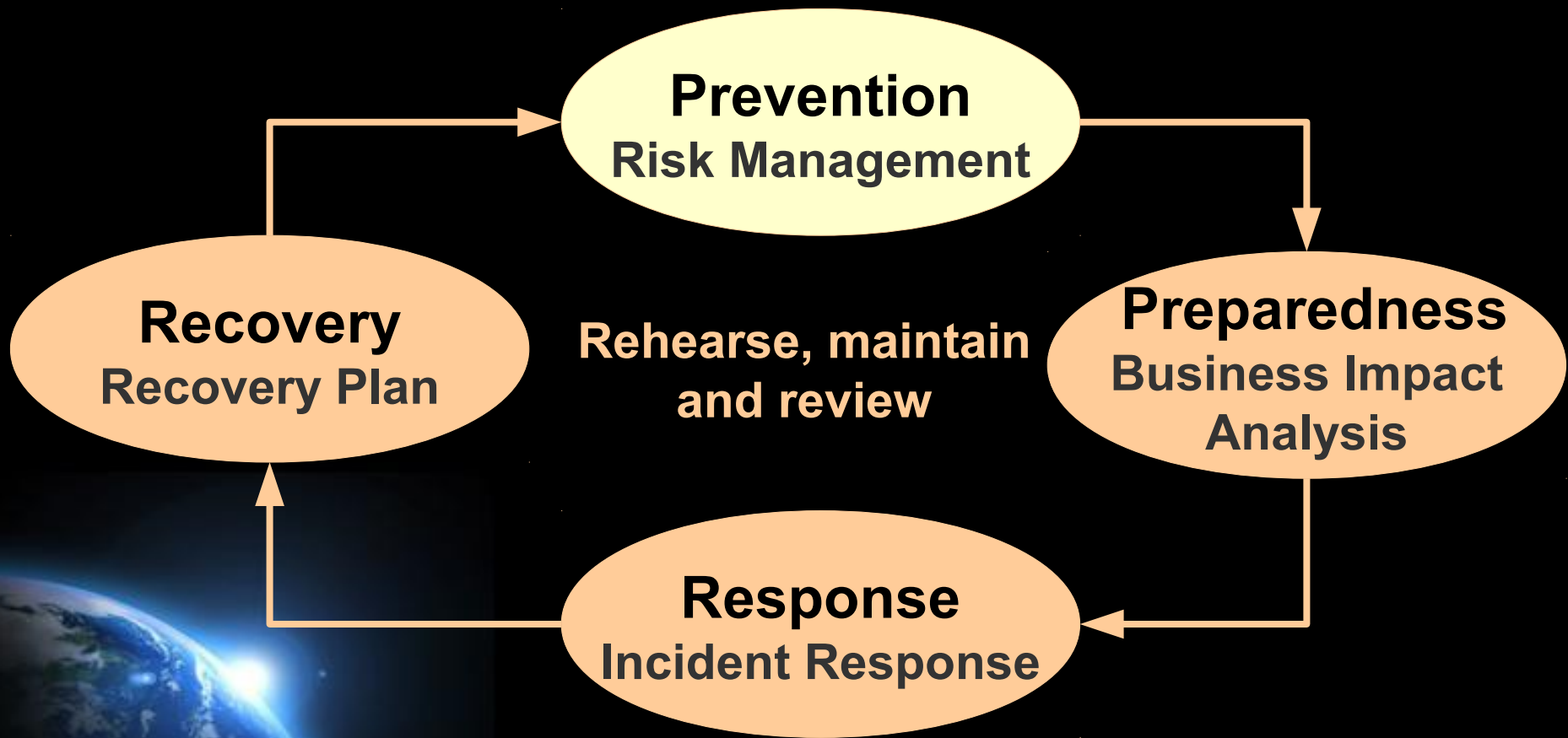
Data from IBM

# Business Continuity Planning

- Goal = minimize service interruption
- Business continuity (BC) specifies how a company **plans** to restore **core business operations** when disasters occur



# Business Continuity Planning Processes



# Prevention Risk Management

- Evaluate Risk
  - Step 1: identify risks that could impact your business
  - Step 2: analyze risks to assess their impacts
  - Step 3: prioritize risks
  - Step 4: treat risks to minimize their impact
  - Step 5: develop and review your risk management plan



# Prevention

## Risk Monitoring

- As business change the risks change accordingly : **periodic review required**
- Monitor and review the strategies to manage the risks : update!
- Why?
  - Reducing insurance fees
  - Reducing the time when business is unable to operate
  - Reducing loss, damage to equipment in general



# Prevention is a lot more...

**Quality control**

**Back-up of data (incl. off site)**

**Staff training**

**Select the proper staff**

**Workplace health & safety**

**Cloud Computing**

**Security measurements**

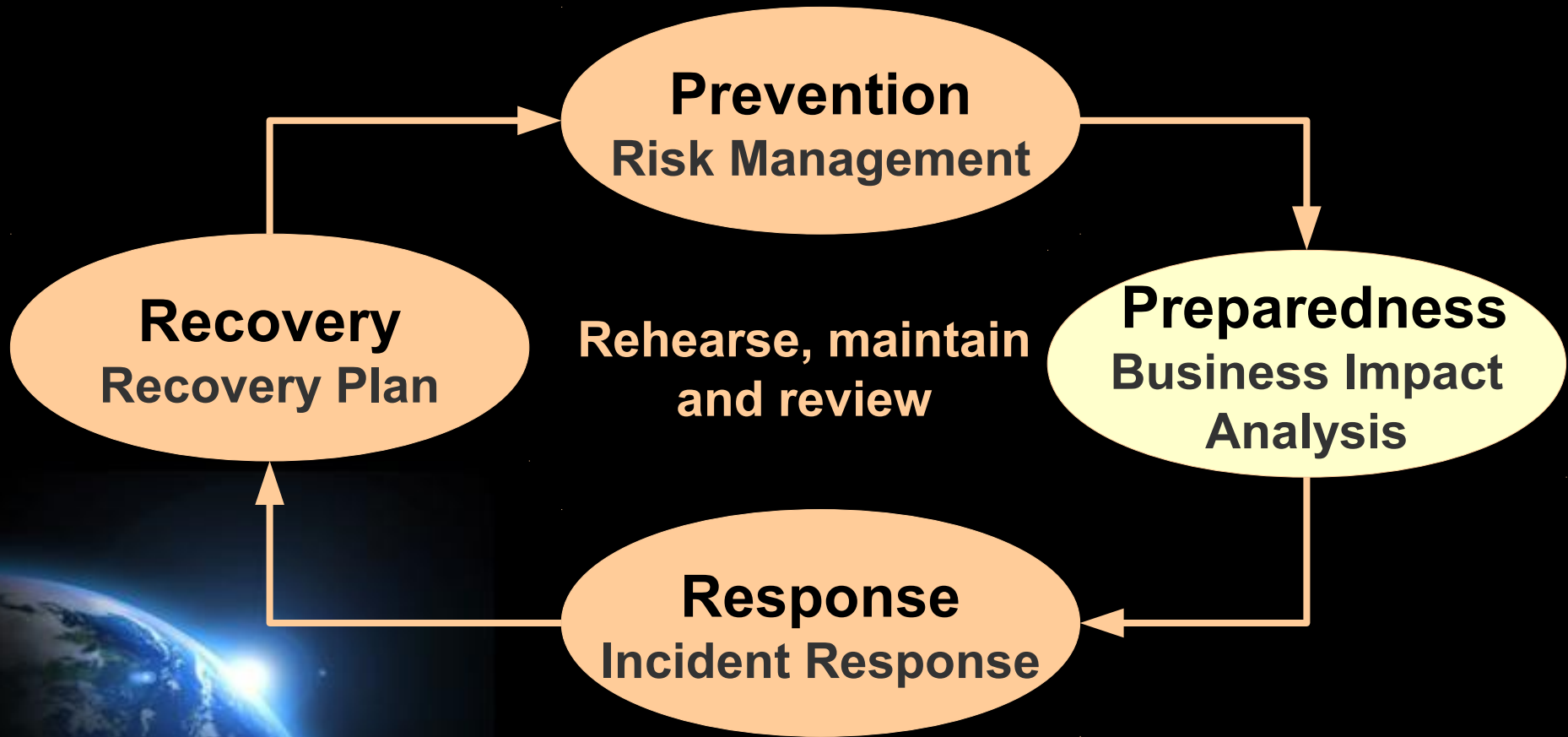
**Emergency procedures**

**Maintenance of HW/SW**

**Evacuation plans**

**Regular drills & tests**

# Business Continuity Planning Processes



# Preparedness what can you do?

- Taking action prior to an incident occurring to ensure an effective response and recovery
- Proactive and planning are key!
- It won't happen to me? *Right....*
- Business Impact Analysis (BIA)



# Preparedness

## Business Impact Analysis

- Discover which processes are vital
- Prioritize and cut scope
  - False assumptions about criticality
  - Understand why certain function are more critical then others
  - At what cost? Management must decide
- Requires cross-departmental collaboration



# Preparedness

## Business Impact Analysis

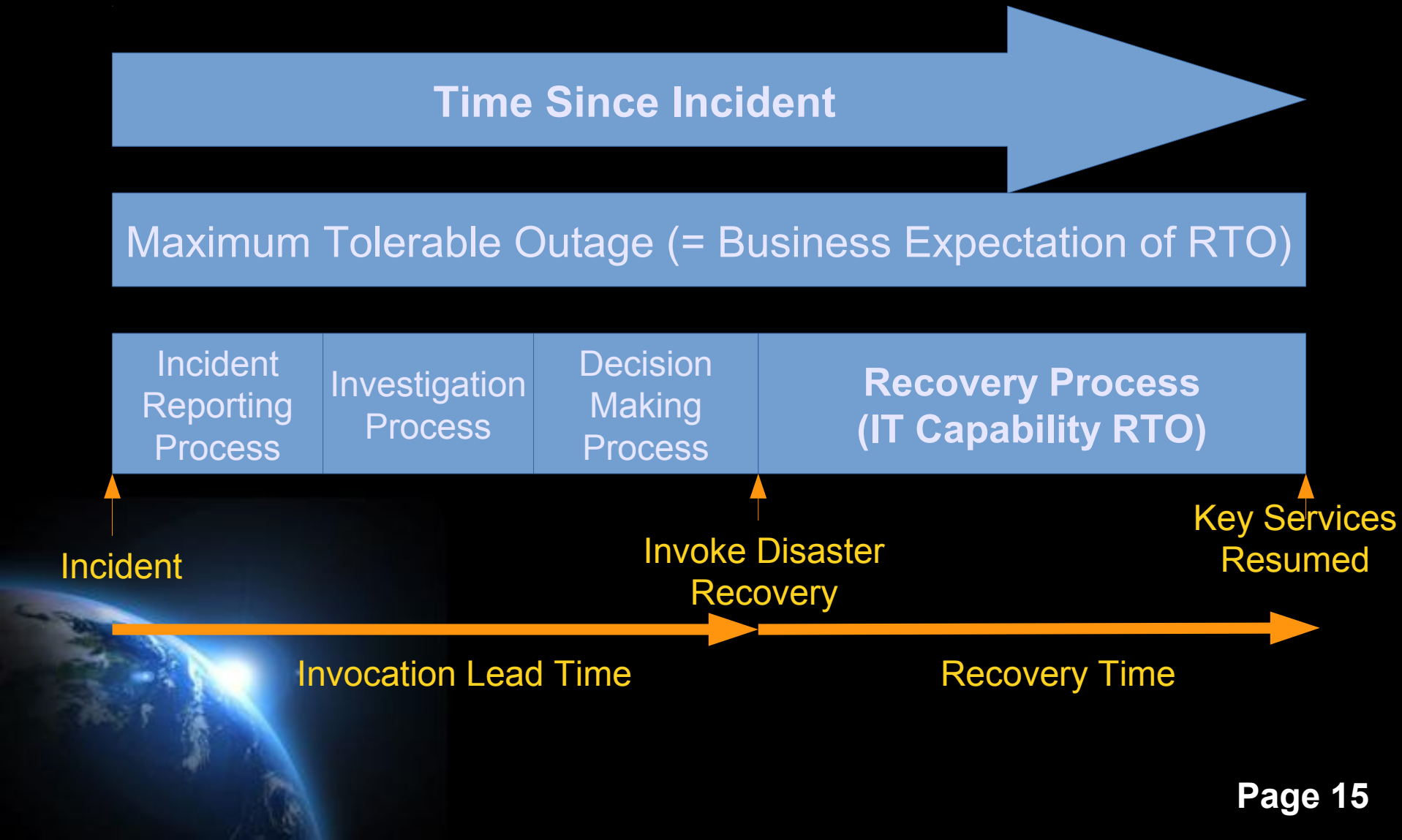
- Executive staff defines mission critical applications
  - Agree upon what is acceptable downtime (Recovery Time Objective or **RTO**)
  - Agree upon your recovery point objective (Recovery Point Objective or **RPO**)
  - Zero downtime is an utopia and becomes extremely expensive
  - You need to guide the executives

# Business Impact Analysis

## RPO - RTO

- **RPO**: Recovery Point Objective
  - How much data are we prepared to lose
    - Zero data loss?
    - Weekly, daily, or in between backups?
- **RTO**: Recovery Time Objective
  - How much downtime can we afford?
  - Lower recovery time = higher cost
  - Focus on critical processes

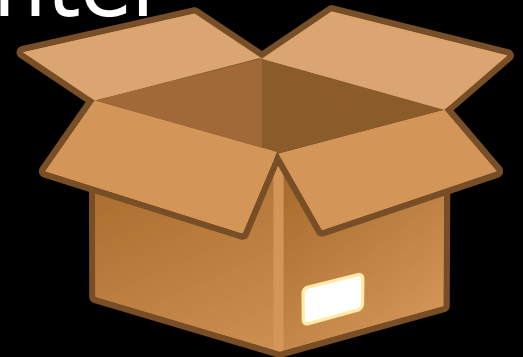
# Business Impact Analysis MTO



# **Business Impact Analysis**

## **Example BIA**

- Company X is a mail order retailer
- Almost all revenue is from on-line sales
- Online catalog of 25.000 items
- Online community message board
- Office in one location, including warehouse, IT and call center





# **Business Impact Analysis Potential Risks**

- Theft (internal & external)
- Fires, Floods, Earthquakes
- Power Outage
- Server Crash (HW or SW)
- Loss of key personnel
- DDoS or web site hacked/outage
- Water pipe burst



# **Business Impact Analysis**

## **Identify Key Processes**

- Pending order
- Tracking stock
- Online assistance
- Credit card processing
- Online message board
- Search database
- IT maintenance

# **Business Impact Analysis Inter-dependencies**

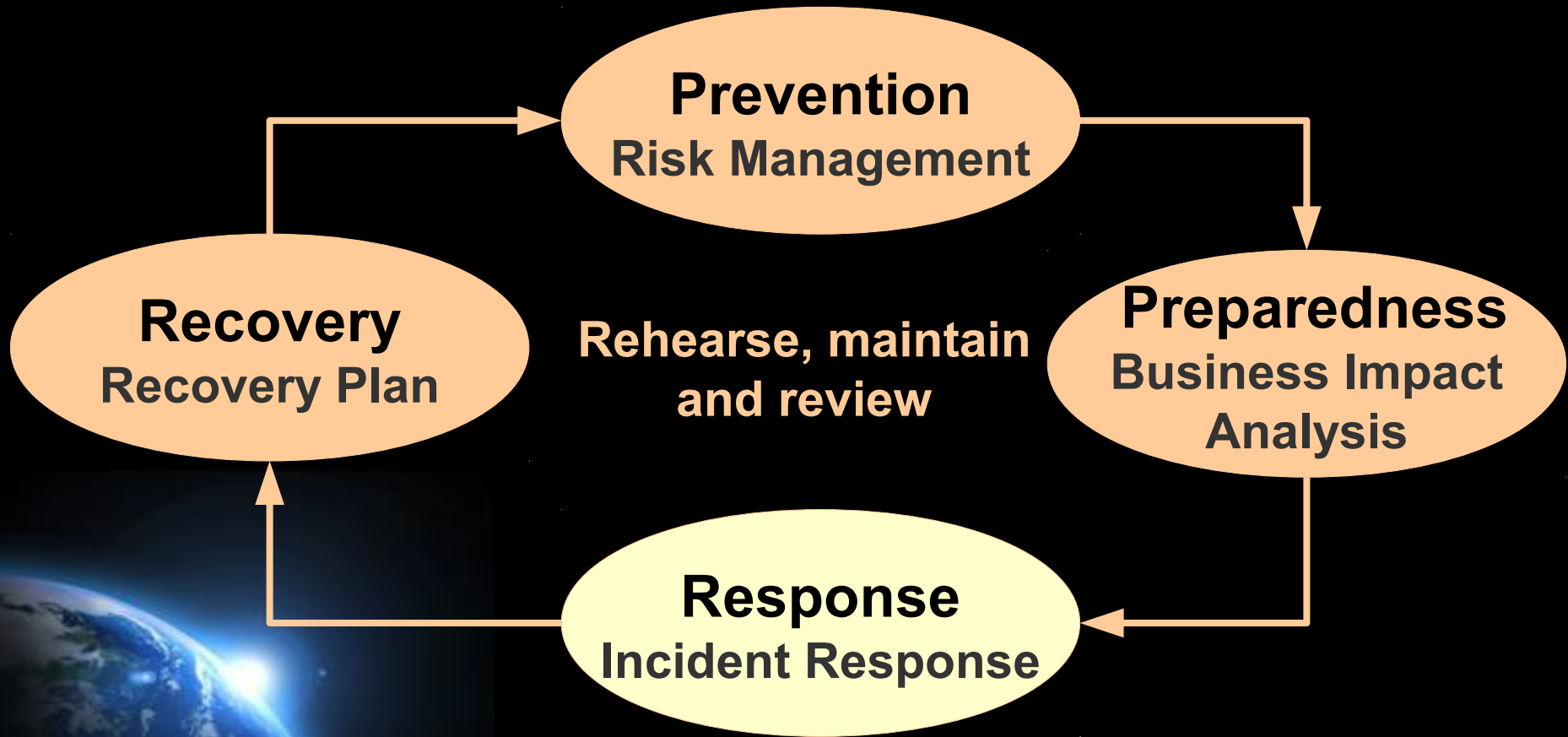
- Pending orders
- Tracking stock
- Online assistance
- Credit card proc.
- Online message board
- Search database
- Stock refill
- IT maintenance
- Public relations
- Legal compliance
- Vendor agreements
- Server room clean-up
- Budgeting
- Emergency loan
- Financial reporting

# Business Impact Analysis

## Impact on operations

- Online store: high
- Credit card processing: high
- Rebuilding: medium
- Message board: low
- Product search: low
- Financial reporting: low
- Based on revenue value!

# Business Continuity Planning Processes



# Response

- A response team: experts who are able to understand and evaluate the specific crisis
  - Team leader
  - Response team
  - Spokesperson
  - Others...
- The crisis should be their only concern!



# Incidence Response Plan

- Scenario's
- Contact list
- Check lists
- A 'GO' pack:
  - Laptop with all docs; Google doc, ...
  - Event logging
  - Evacuation plan
  - Authority list



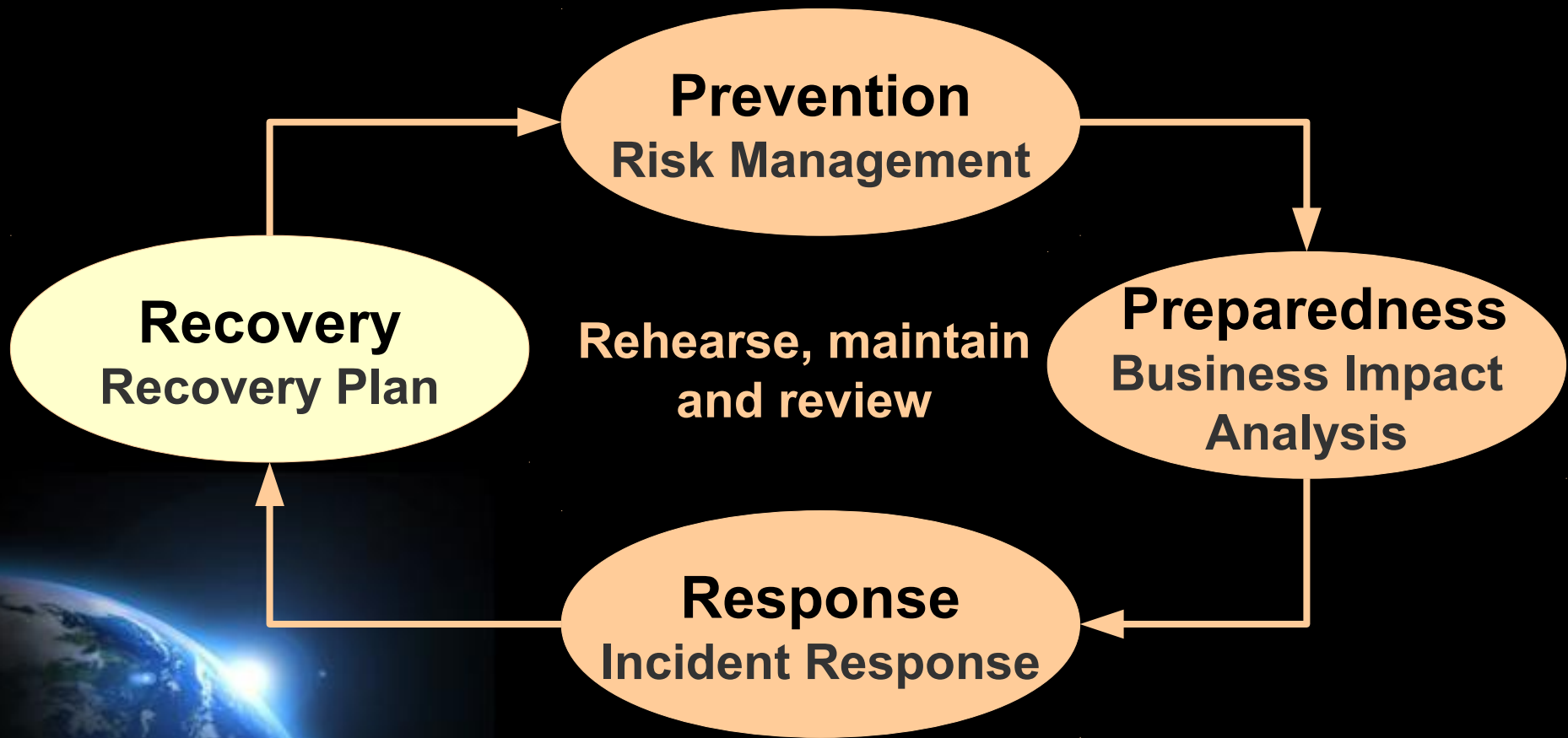
# Activate the response team

- Define the real problem and lay out the strategy to resolve
- Act quickly and do not forget to communicate (twitter,...)
- Assume the worst so you are prepared (escalation)
- Use the Subject Matters Experts effectively





# Business Continuity Planning Processes



# Recovery

- During the incident we concentrate on disaster recovery
  - We have a DR plan, right?
- What about after the crisis?
  - Damage to property, IT equipment, ...
  - Damage to reputation (do not forget communication)
  - Insurance is very important



# Disaster Recovery

Disaster Recovery (DR) is the process, policies and procedures that are related to preparing for recovery or continuation of **technology infrastructure** which are vital to an organization after a natural or human induced crisis

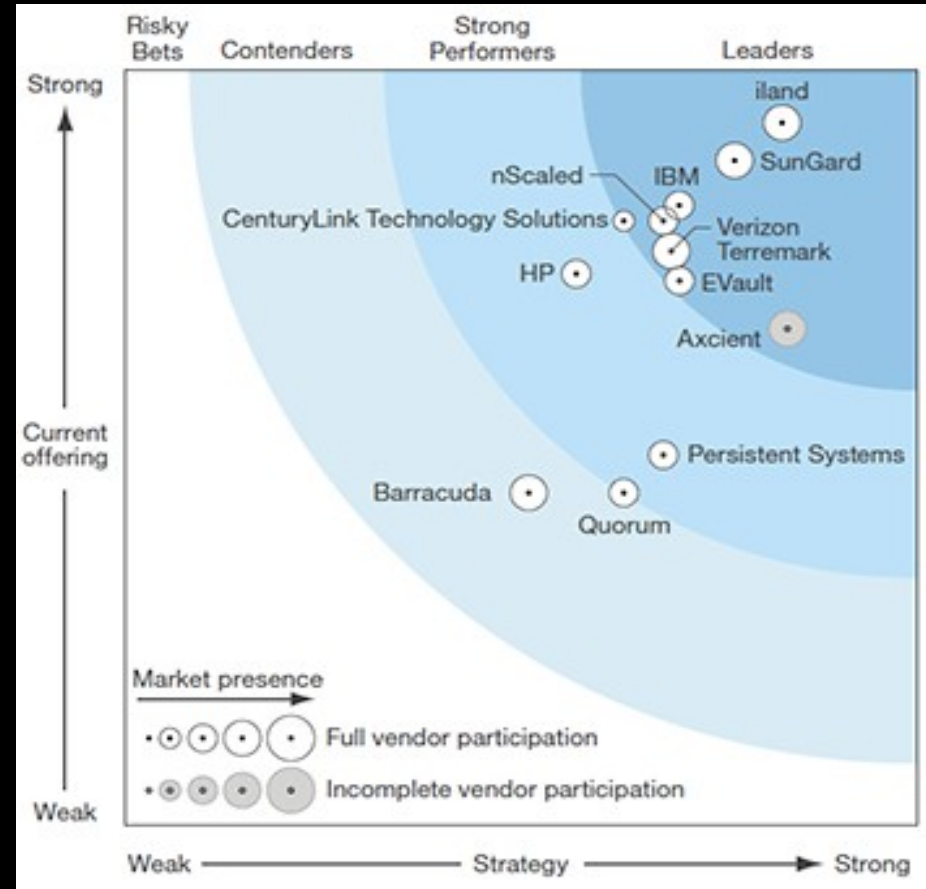


# Disaster Recovery is not

- Backup, which is mostly about data loss prevention, DR is about *service availability* (low RPO and RTO)
- Data replication to ensure consistency between redundant sites
- DR complements other *High Availability* activities (dealing with DR prevention), DR is for the times when prevention fails

# Cloud Disaster Recovery

- Approaches
  - Do it yourself
  - DRaaS
- Techniques
  - Cold DR
  - Warm DR
  - Hot DR



Source: The Forrester Wave:  
DraaS Providers, Q1 2014

# Critical server: backup system

- In DR site the backup server is key
- Do not forget to create a DR plan for this backup server (rear)
- Synchronize the backup data
  - Disk synchronization
  - Tapes (vaulting)
  - Cloud Storage (if size permits)



# Bareos Backup Server

- Bareos (Backup Archiving REcovery Open Sourced) is a fork of Bacula (2010)
- URL: <http://www.bareos.org/en/>
- Is an excellent choice as it works with
  - Tapes, disks, deduplication
  - Cloud storage
  - Integration with rear



# Relax-and-Recover (rear)

- Open Source Bare Metal Restore (DR)
- URL: <http://relax-and-recover.org/>
- Online: snapshot of running system
  - Creates bootable image (ISO, PXE, USB)
  - Creates archive via GNU tar, rsync, or
  - Integrates with backup software:
    - Bareos, bacula, rbme, duplicity (open source)
    - Commercial backup software (TSM, NBU, DP, GALAXY, NSR, SESAM)



# Rear features

- Fully automated recovery
  - On same hardware
  - Similar hardware
- P2V, V2V, V2P, migrate storage (SAN)
- During recovery rear will
  - Prepare storage (partitioning, file systems, mount points)
  - Restore archive from backup
  - Install boot loader

# Rear using bareos

- Clients system of bareos
  - BACKUP=BAREOS
  - Rear -v mkrescue
- Bareos backup server
  - BACKUP=NETFS
  - BACKUP\_URL=<external storage>, e.g. NFS, USB, ISO
  - OUTPUT=ISO, USB

# Recover system

- Recover bareos client with rear
  - Rear -v recover
  - Recreates all file systems
  - Uses bareos to restore all data
- Recover bareos server with rear
  - Rear -v recover
  - Recreates all file systems
  - Uses external storage to restore data
  - Restore latest backup via tape(s)



# Business Continuity



# At your service...



<http://www.it3.be/rear-support>